Hybrid Knowledge-Graph-Enhanced Retrieval-Augmented Generation for Academic Information Systems

Second Review

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Abstract

Academic data is typically unstructured and fragmented across institutional websites, limiting the effectiveness of keyword-based and embedding-only retrieval systems. Hybrid knowledge-graph-enhanced retrieval-augmented generation improves contextual grounding and relational reasoning by combining semantic vector retrieval with structured knowledge graph inference, supporting SDG 4 and SDG 9 [1, 2].

Introduction

- Growing dependence on online academic information
- Institutional data scattered across multiple web sources
- Limited relational reasoning in conventional retrieval systems
- Need for intelligent, knowledge-driven academic information systems

Literature Review

Sl.No	Title	Methodology	Results	Limitations
1	Knowledge graph-extended RAG for QA [1]	KG-guided retrieval + LLM	Improved multi-hop accuracy	Requires structured knowledge graph
2	RAG for educational applications [2]	Semantic retrieval + LLM	Reduced hallucination	Dependent on retriever quality

Research Gaps Identified

- Limited use of knowledge graphs in academic RAG systems
- Lack of hybrid vector-graph retrieval pipelines
- Insufficient explainability in LLM-based academic QA

Problem Statement

To develop a hybrid knowledge-graph-enhanced retrieval-augmented generation system for accurate and explainable academic information access.

Objectives

1. To collect academic data from institutional websites using web scraping techniques. ✓
2. To convert the collected text data into vector embeddings for semantic search and retrieval.✓
3. To construct a knowledge graph of academic entities for Mar Baselios College of Engineering and Technology.
4. To integrate hybrid retrieval for query answering.
5. To improve factual grounding and relational reasoning.

Feasibility Study

Technical Feasibility

- Uses existing academic data from institutional websites
- Integrates proven RAG models with knowledge graph techniques
- Supports semantic and relationship-based information retrieval
- Can be deployed using standard cloud or local server infrastructure

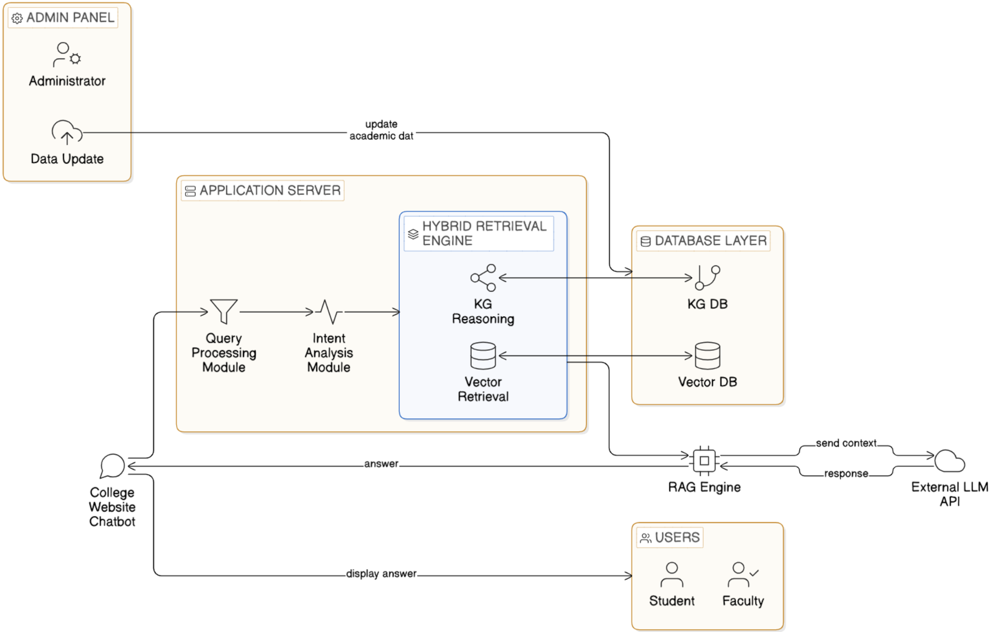
Economic Feasibility

- Relies on open-source NLP, vector databases, and graph frameworks
- No requirement for specialized hardware
- Low development and deployment cost
- Minimal maintenance cost after implementation

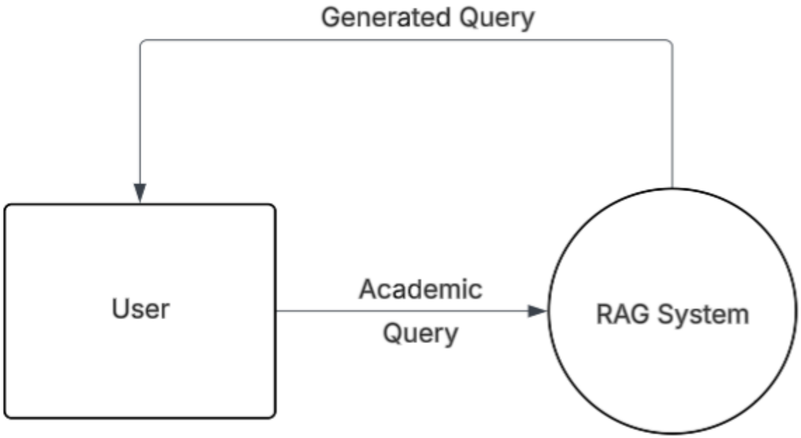
Operational Feasibility

- Provides an easy-to-use chatbot interface for users
- Reduces manual effort for administrative staff
- Requires minimal user training
- Suitable for continuous operation on academic websites

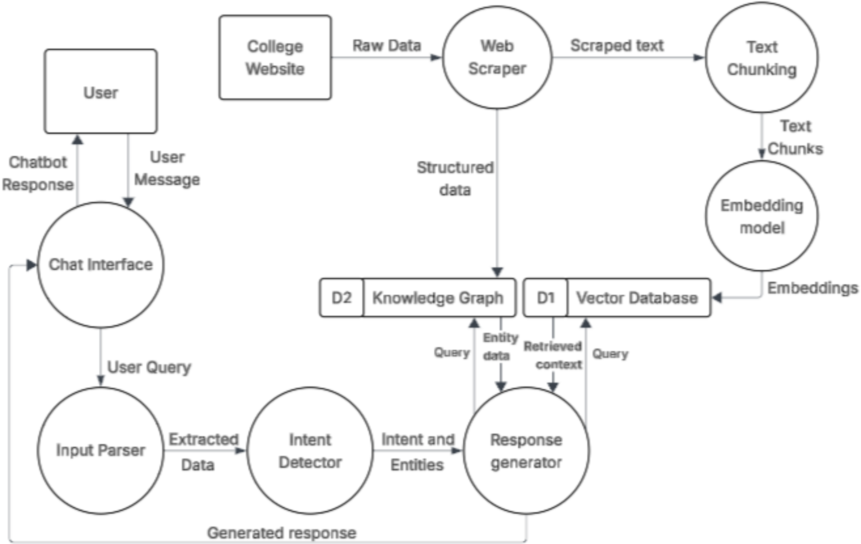
System Architecture



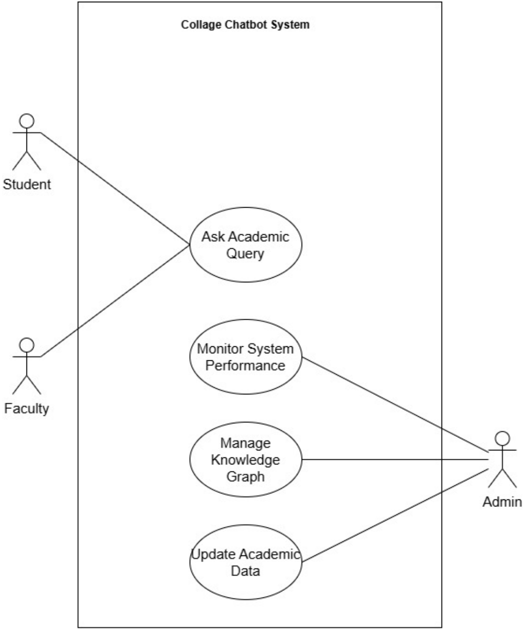
Data Flow Diagram-0



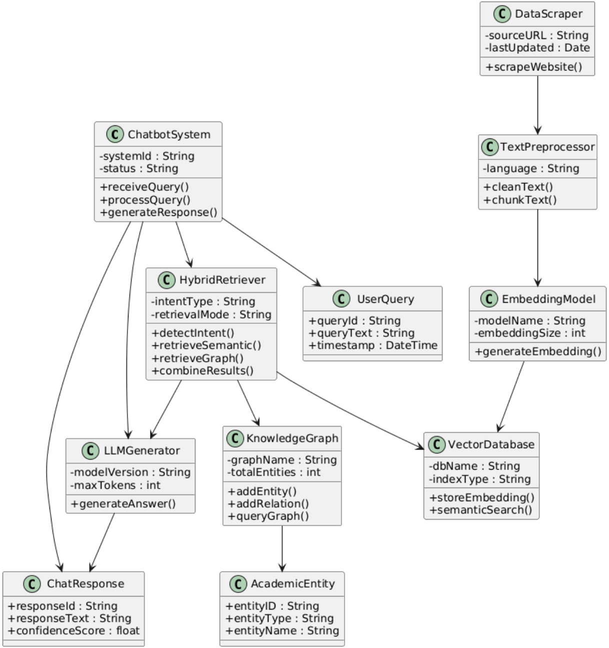
Data Flow Diagram-1



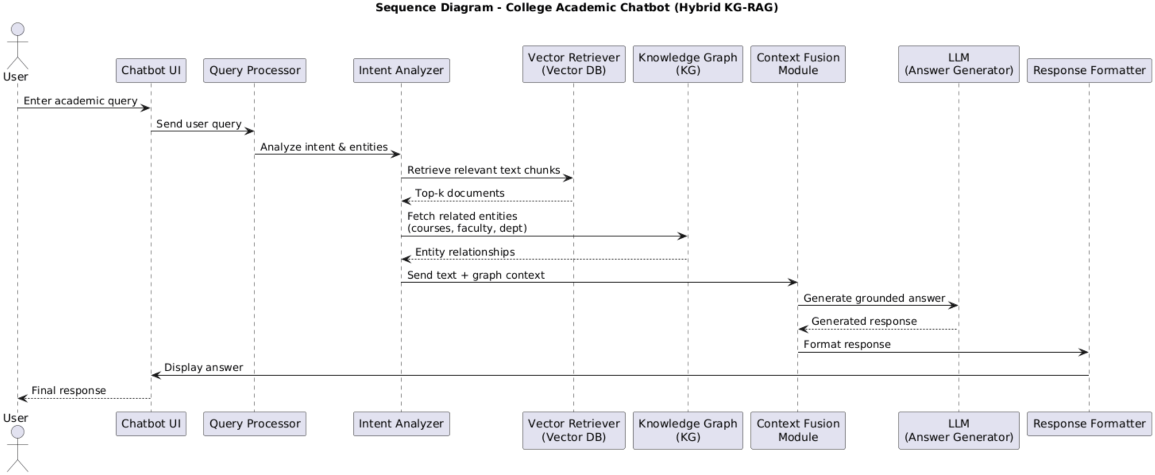
Use Case Diagram



Class Diagram



Sequence Diagram



End User Feedback

- Users mainly visit the college website for exam schedules, syllabus, and academic calendar.
- Many users find it difficult to locate information due to scattered content and complex navigation.
- Common issues include too many clicks and lack of centralized academic information.
- Most users believe a chatbot would help in faster and easier information access.
- Users prefer the chatbot to provide exam-related, academic, and department-specific information.

Methodology

- **Data Collection**
  Academic data such as exam schedules, syllabus, and faculty details are collected from institutional websites using Python libraries like *Requests* and *BeautifulSoup*.
- **Data Preprocessing**
  The collected data is cleaned by removing unnecessary tags and symbols. The text is then divided into smaller chunks for efficient processing using *Regular Expressions* and *Text Splitters*.
- **Vector Embedding Generation**
  Each text chunk is converted into vector embeddings using models such as *SentenceTransformer*, enabling the system to capture the semantic meaning of the text.
- **Vector Database Storage**
  The generated embeddings are stored in a vector database (*FAISS*) to enable fast similarity search and efficient retrieval of relevant documents.

Methodology

- **Knowledge Graph Construction**
  A knowledge graph is created using a *Simple Knowledge Graph* approach to represent relationships between academic entities such as departments, courses, and faculty members.
- **Hybrid Retrieval**
  When a user submits a query, the system retrieves the Top-K relevant documents from the vector database and related entities from the knowledge graph to gather useful information.
- **Answer Generation**
  The retrieved information is combined and passed to a *Large Language Model (LLM)* to generate accurate and context-aware responses.
- **Response Delivery**
  Finally, the generated answer is formatted and displayed to the user through the chatbot interface.

Implementation

Interpretation

- Clear clusters indicate that the embedding model is successfully grouping similar content together.
- Tables (purple) dominate the dataset and appear in multiple clusters, suggesting many structured data entries.
- Profiles (green) form a tight cluster, likely because faculty profiles have similar text structure.
- Sections and regulations appear near each other, showing semantic similarity between policy-related content.
- A few scattered points represent unique or less common content.

Conclusion

- Hybrid KG-RAG improves academic information retrieval
- Supports both semantic and relational queries
- Provides grounded and explainable responses
- Aligned with SDG 4 and SDG 9

References

[1] J. Linders and J. M. Tomczak, “Knowledge graph-extended retrieval augmented generation for question answering,” *Applied Intelligence*, vol. 55, no. 17, pp. 1102–1118, 2025.

[2] Z. Li, Z. Wang, W. Wang, K. Hung, H. Xie, and F. L. Wang, “Retrieval-augmented generation for educational application: A systematic survey,” *Computers and Education: Artificial Intelligence*, vol. 8, p. 100417, 2025.

Thank You